

Week 2 Task Report

Progress Completed:

- Successfully converted my basic Python data listener into a fully working ROS 2 Node using the `rc_lpy` library.
- Created a proper ROS 2 package named `mobile_imu_control` and configured the build files (`setup.py` and `package.xml`).
- Wrote the math to convert the raw accelerometer data (`ax`, `ay`, `az`) from the phone into exact tilt angles (Pitch and Roll).
- Mapped these tilt angles to a ROS `Twist` message. The phone's pitch now controls the forward/backward speed, and the roll controls the left/right turning.
- Successfully achieved real-time control of the TurtleBot3 in Gazebo using my mobile phone as a wireless steering wheel!

Challenges Faced:

- **Axis Mismatch (Robot moving wrong way):** When I first ran the node, the robot moved backward when I tilted forward, and turned when it was supposed to go straight. I found out that Android phones have a different X/Y axis layout than standard robots. I solved this by customizing the Python math (`math.atan2`) to swap and invert the axes so the robot responds naturally to human hand movements.

Plans for the Next Week (Week 3):

- **Implement Sensor Filtering:** Right now, the robot shakes slightly because the phone's raw sensor is very sensitive. Next week, I will code a "Simple Moving Average" filter to smooth out the data.
- **Implement a Deadzone:** I will add code so the robot completely stops moving when the phone is held flat.

Demo

<https://github.com/ras-al/Mobile-IMU-Telemetry-Controller-for-TurtleBot3/blob/main/README.md>